

The Role of Ethics in AI Development

WHY THIS MODULE MATTERS

Reputation is built over years and lost in days. This module establishes the ethical foundation for every AI decision your organisation makes — what it stands for, and what responsible AI looks like in practice, not just in policy.

2.1 Ethics Is a Design Input, Not an Afterthought

A common misconception treats ethics as a review step that happens after a system is built — a checklist applied just before launch. In practice, the decisions with the greatest ethical weight are made much earlier: what problem the system is built to solve, what data trains it, what trade-offs are acceptable between accuracy and fairness, and who is consulted before deployment. By the time a system reaches a late-stage review, many of its most consequential properties are already fixed.

Treating ethics as a design input means asking evaluative questions at the same points where technical questions are asked: during problem framing, data collection, model selection and evaluation design. It does not require every engineer to become an ethicist; it requires teams to build the habit of asking who could be harmed by this system, and how would we know.

CASE STUDY — AMAZON'S ABANDONED HIRING TOOL

Between 2014 and 2017, Amazon built an internal AI tool to rank job applicants' CVs, trained on ten years of the company's own hiring data. Because that historical data came from a workforce that was mostly male, particularly in technical roles, the model taught itself that male candidates were preferable, and penalised CVs containing the word “women's” (as in “women's chess club captain”) and downgraded graduates of two all-women's colleges. Engineers edited the model to stop it penalising those specific terms, but could not be confident it had not learned other, subtler proxies for gender from the same biased training data. Reuters reported in 2018 that Amazon had scrapped the tool entirely rather than risk deploying it. No one at Amazon set out to build a sexist hiring tool — the bias was inherited entirely from historical outcomes the model was trained to reproduce.

2.2 Core Ethical Principles Applied to AI

Several recurring principles appear across most professional and regulatory frameworks for AI ethics, including the UK Government's five cross-sectoral principles referenced in Module 1. They are not a checklist to complete once, but lenses to apply throughout a system's lifecycle.

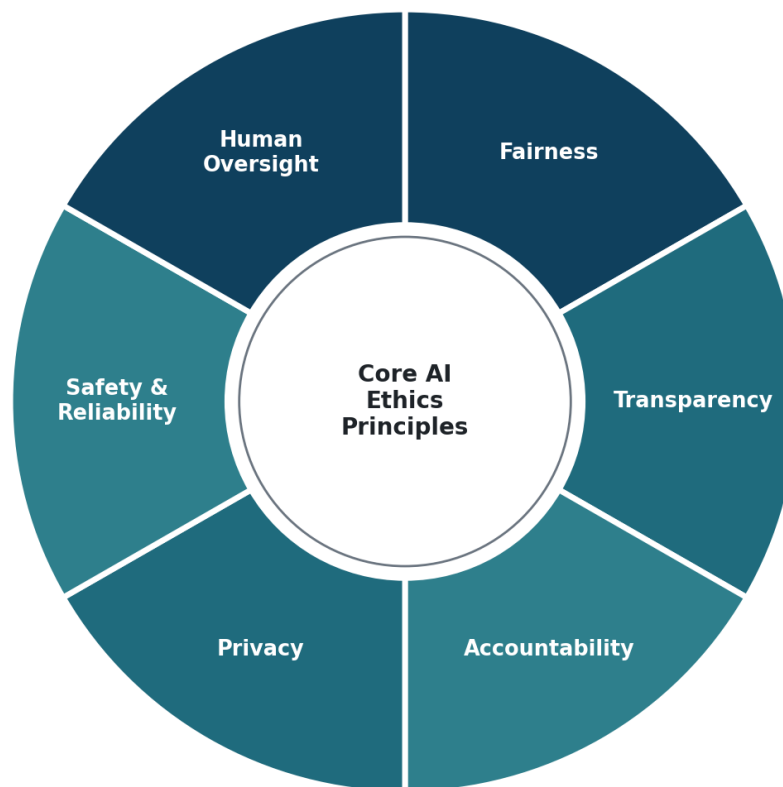


Figure 2.1 — Six core AI ethics principles referenced throughout this course.

- **Fairness** — the system's outcomes should not systematically disadvantage people based on protected or proxy characteristics, and disparities should be identified and justified or corrected.
- **Transparency** — people affected by a system should be able to understand, at an appropriate level, that an automated system is involved and broadly how it works.
- **Accountability** — a specific person or function must be answerable for a system's outcomes; responsibility should never be diffused to the point that no one owns it.
- **Privacy** — data used to train and operate a system should be collected, stored and used consistently with UK data protection law, covered fully in Module 6.
- **Safety and reliability** — a system should behave predictably within its intended scope, and degrade gracefully, or fail safely, outside it.

— **Human oversight** — consequential decisions should retain a meaningful point of human review or intervention, proportional to the stakes involved.

2.3 Three Competing Definitions of Fairness

“Make the system fair” sounds like a single instruction, but fairness in AI has several mathematically distinct, and sometimes mutually incompatible, definitions. Understanding the difference matters, because a system can satisfy one definition of fairness while clearly failing another — and stakeholders who disagree about which definition matters most will disagree about whether the same system is fair, using the same numbers.

— **Demographic parity** — the system should flag or select people from different groups at similar overall rates, regardless of other factors.

— **Error-rate balance (equalised odds)** — the system's false positive rate and false negative rate should be similar across groups, so that being wrongly flagged, or wrongly cleared, is equally likely regardless of group membership.

— **Calibration (predictive parity)** — among people who receive the same score or risk level, the actual outcome rate should be similar across groups, so the score means the same thing whoever receives it.

It is a mathematical fact, not a design failure, that when the underlying base rates differ between groups, a system generally cannot satisfy error-rate balance and calibration simultaneously. Choosing which definition to prioritise is therefore an ethical decision, not a purely technical one — and it is a decision that should be made deliberately, and documented, rather than left to whichever definition the development team happened to test against.

CASE STUDY — COMPAS AND THE FAIRNESS TRADE-OFF

In May 2016, the investigative newsroom ProPublica published an analysis of COMPAS, a recidivism risk-scoring tool used by US courts, based on over 7,000 defendants scored in Broward County, Florida. It found that among defendants who did not go on to reoffend, Black defendants were flagged “high risk” at a false positive rate of roughly 45%, compared with roughly 24% for white defendants — nearly double. By this error-rate measure, ProPublica concluded COMPAS was biased against Black defendants.

Northpointe, the company that built COMPAS, countered that the tool was calibrated: among defendants who received the same risk score, actual reoffending rates were nearly identical across race — for example, roughly 60% of white and 61% of Black defendants scored a 7 went on to reoffend. By this second measure, the company argued COMPAS was fair. Both sides were using real numbers from the same dataset; they were simply applying different, both legitimate, definitions of fairness — which is exactly why this case remains the standard teaching example for this trade-off.

TRY THIS — APPLY THE THREE DEFINITIONS

A different risk-scoring tool used by a UK lender flags loan applicants as “high risk” or “low risk”. Among applicants who did not default, Group A is flagged high risk 10% of the time and Group B is flagged high risk 30% of the time. Among applicants who scored “high risk”, 40% of Group A defaulted and 40% of Group B defaulted. Work through the three definitions above: does this system satisfy error-rate balance? Does it satisfy calibration? Does it satisfy demographic parity? (Answer: it satisfies calibration, since the same score means the same default rate for both groups, but clearly fails error-rate balance, since Group B is falsely flagged three times as often — the same tension COMPAS illustrates.)

2.4 Fairness and the Equality Act 2010

In the UK, fairness in AI is not only good practice — it intersects directly with the Equality Act 2010, which prohibits discrimination on the basis of protected characteristics including age, disability, race, sex, sexual orientation, religion or belief, and others. An AI system that produces systematically worse outcomes for people with a protected characteristic, even unintentionally, can amount to indirect discrimination unless the organisation can objectively justify it. This is a legal as well as an ethical constraint on system design, and it was precisely the risk Amazon's engineers judged too great to accept when they scrapped their hiring tool.

2.5 Why Good Intentions Are Not Enough

Neither Amazon nor Northpointe set out to build a discriminatory system. Both cases above show harm arising from reasonable-seeming decisions — training on convenient historical data, choosing one defensible fairness metric over another — compounding in ways no individual decision-maker fully saw in advance. This is why ethics in AI development functions less like a personal virtue and more like an engineering discipline: it requires structures that catch what good intentions alone will miss.

The Value-Alignment Gap

Organisations frequently state values — fairness, trust, user wellbeing — that are not reflected in the metrics their teams are actually measured against. A team incentivised purely on engagement or conversion will optimise for those numbers even if it privately holds broader values. Closing this gap means translating stated values into the metrics, review gates and incentives that actually shape day-to-day decisions — deciding, and documenting, which fairness definition from section 2.3 actually governs sign-off for a given system, rather than leaving it implicit.

2.6 Ethics in Physical AI Infrastructure Roles

The principles above are usually discussed in relation to the AI systems being built. They apply equally to the AI-enabled systems used to run the buildings those systems live in. Modern data centres increasingly rely on AI themselves — facial recognition and biometric access control at entry points, machine-learning-driven anomaly detection on CCTV feeds, and predictive models that manage cooling and power load. Staff who operate or are subject to these systems face the same ethical questions as any other AI deployment.

- **Consent and necessity** — is biometric access control the least intrusive way to achieve the required security outcome, and have staff and visitors been properly informed?
- **Accuracy and fairness** — facial recognition systems have documented, well-evidenced accuracy gaps across skin tone and gender; a failed match at a secure door is not a neutral inconvenience if it falls disproportionately on some staff more than others.
- **Surveillance proportionality** — anomaly-detection systems that flag “unusual” staff behaviour should be calibrated and reviewed so that ordinary variation in how people work is not treated as a security event.
- **Whistleblowing and escalation** — staff who notice a safety, security or ethical problem with a system need a route to raise it that does not depend on going around their own line management.

These are not hypothetical concerns for a recruitment or facilities audience — they are day-to-day operational questions, and Module 9 returns to them with a specific focus on the data centre environment.

KEY TAKEAWAYS

- Ethical consideration is most effective when it happens during design, not only before launch — Amazon's hiring tool was already biased by the time anyone tested it for fairness.
- Fairness has at least three distinct, sometimes mutually incompatible mathematical definitions: demographic parity, error-rate balance and calibration.
- The COMPAS case shows that two parties can each be technically correct about “fairness” while reaching opposite conclusions, because they are measuring different things.
- In the UK, unfair AI outcomes linked to protected characteristics can engage the Equality Act 2010, not only internal ethics policy.
- The same ethical principles apply to the AI-enabled systems — biometric access, CCTV analytics — that run data centres, not only to the AI products those data centres host.

FURTHER READING

- Reuters — Dastin, J., “Amazon scraps secret AI recruiting tool that showed bias against women” (2018)
- ProPublica — “Machine Bias”, Angwin et al. (2016), propublica.org
- Equality and Human Rights Commission — guidance on the Equality Act 2010

MODULE 2 CHECK-IN QUIZ

Five questions to confirm your understanding before moving on. Answers are provided in the Answer Key at the end of the course.

1. Why does this course argue that ethics should be treated as a design input?

- A. Because late-stage review is more effective than early design decisions
- B. Because many consequential properties of a system are already fixed by the time of late-stage review
- C. Because ethics only matters after deployment
- D. Because design decisions are always ethically neutral

2. Which UK legislation is most directly engaged when an AI system produces systematically worse outcomes for people with a protected characteristic?

- A. The Data Protection Act 2018
- B. The Equality Act 2010
- C. The Data (Use and Access) Act 2025
- D. The Human Rights Act 1998 alone

3. In the postcode-pricing example, why could the model be problematic even without using a protected characteristic directly?

- A. Postcode is always illegal to use in pricing
- B. Postcode can act as a proxy correlated with protected characteristics such as ethnicity or income
- C. Postcode is classified as special category data
- D. Insurance pricing is exempt from fairness requirements

4. What does this course mean by the “value-alignment gap”?

- A. A technical gap between model versions
- B. A mismatch between an organisation's stated values and the metrics teams are actually measured against
- C. A gap in data protection law coverage
- D. The difference between UK and EU AI regulation

5. Which of the following is NOT one of the six core ethical principles introduced in Module 2?

- A. Fairness
- B. Transparency
- C. Market competitiveness
- D. Human oversight